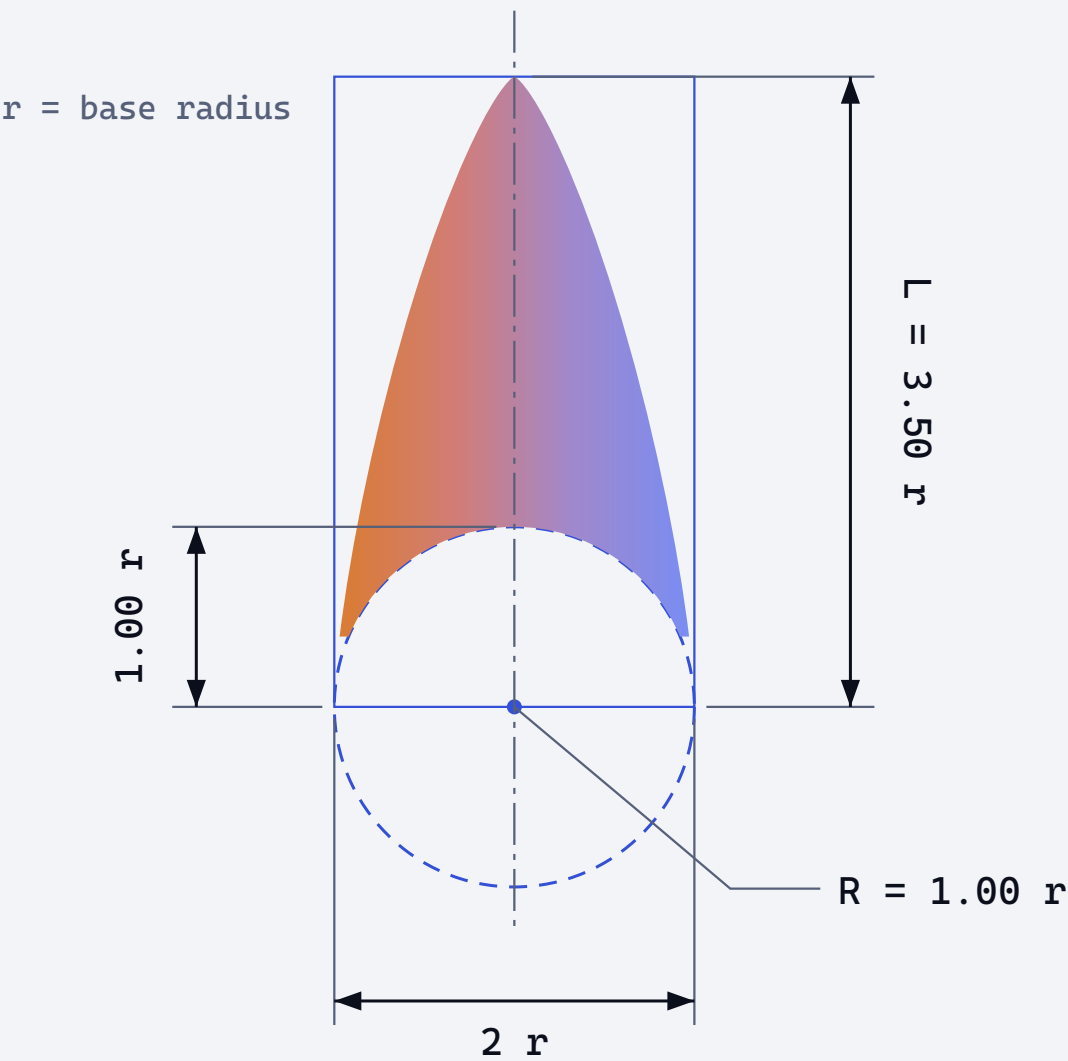
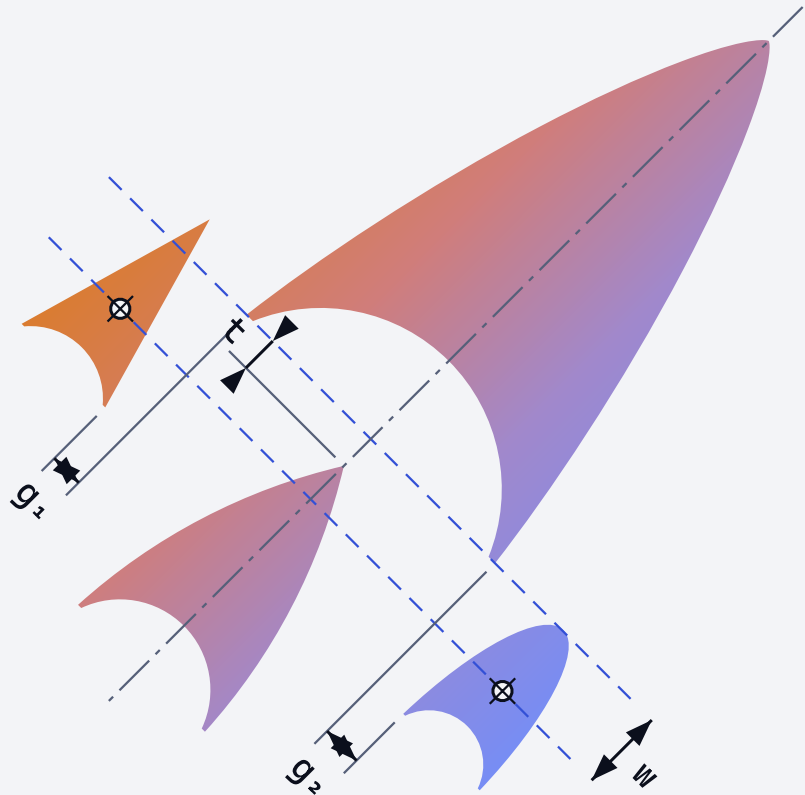


1 · CONE (VON KÁRMÁN)

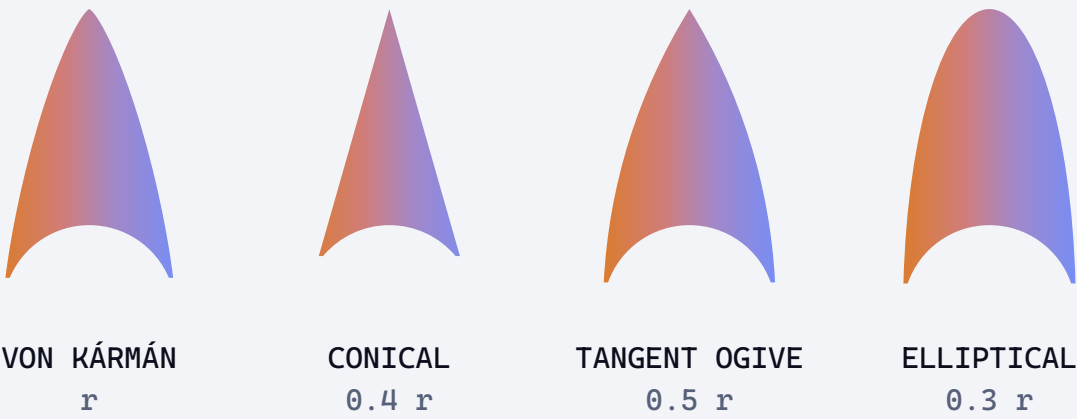


2 · SPACING ON THE 45° GRID

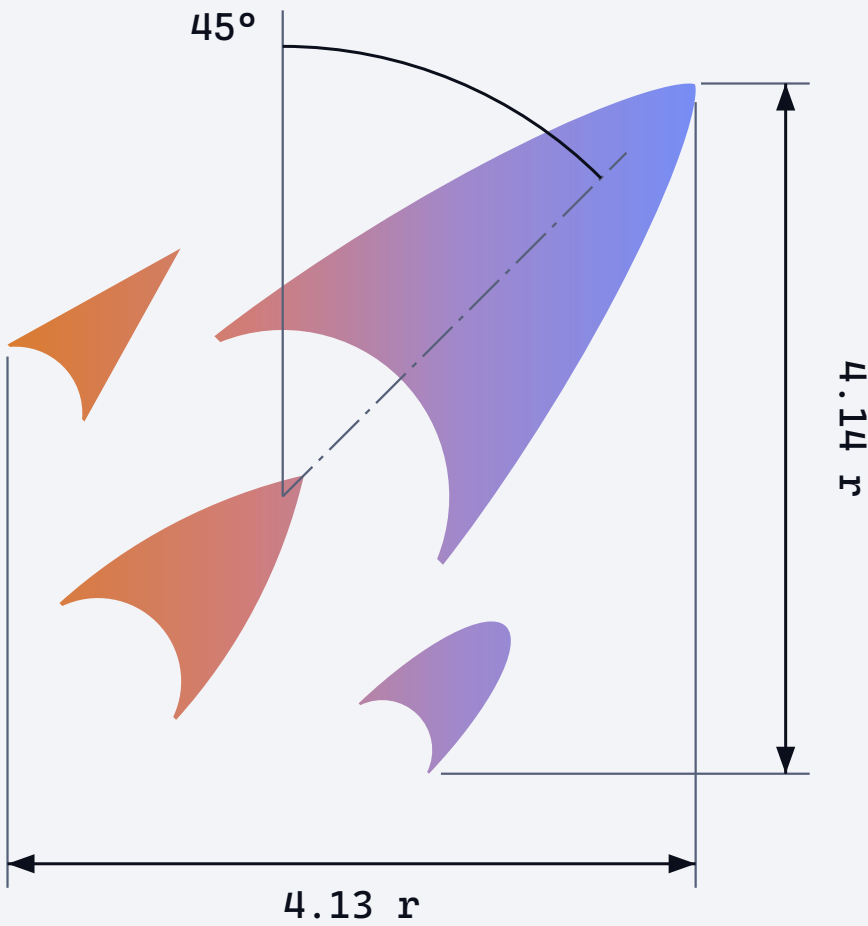


$g_1 = 0.19 r$, $g_2 = 0.23 r$ from the Von Kármán foot corners to the wings. Wing centroids share a line $w = 0.47 r$ below the Von Kármán foot line. The ogive is the tail, on the same axis, tip $t = 0.21 r$ below the Von Kármán foot line. Lay out on the grid, then the grid leans as one piece.

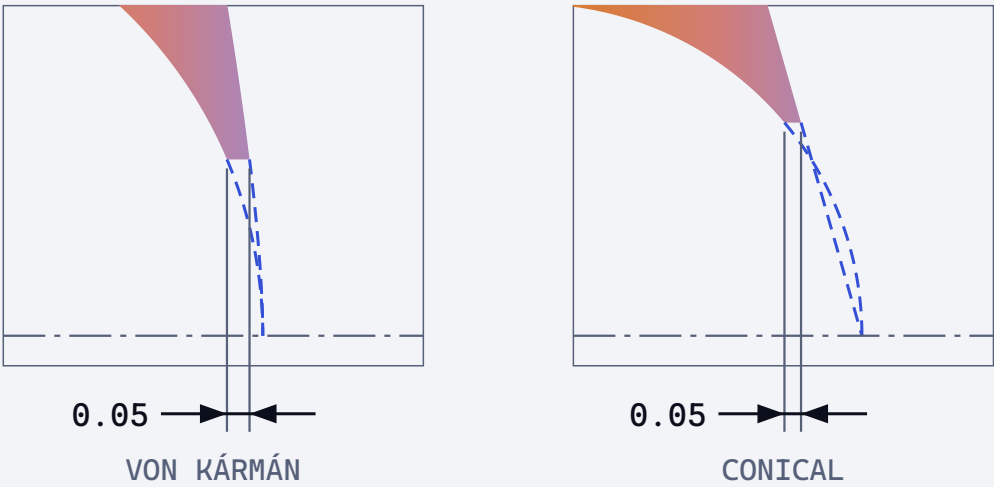
4 · PROFILES (EQUAL L / R)



3 · CLUSTER



5 · FEET (DETAIL)



Each foot is cut flat, parallel to the base (dash-dot), where the wall between flank and notch is $0.05 \times$ the cone's base radius wide. Dashed: the uncut outline. On the conical cone the notch arc runs outside the straight flank near the base, so the cut also removes that sliver.

CONE	PROFILE	RADIUS	BASE CENTER (u, v)
MAIN	VON KÁRMÁN	r	(+0.0, +0.0)
WEST	CONICAL	$0.4 r$	(-1.5, +0.8)
EAST	ELLIPTICAL	$0.3 r$	(+1.5, +0.7)
SOUTH	TANGENT OGIVE	$0.5 r$	(+0.0, +1.6)

u, v in r on the leaned grid, v down.

Standard nose-cone profiles, $L = 3.50 \times$ base radius, notch 1.00 deep, feet cut flat where the wall is 0.05 wide (all $\times$ base radius). Von Kármán is the LD-Haack series ($C = 0$), drawn as smooth curves. The whole cluster leans 45° as one piece. One gradient spans the whole cluster, M orange left to 0 blue right.